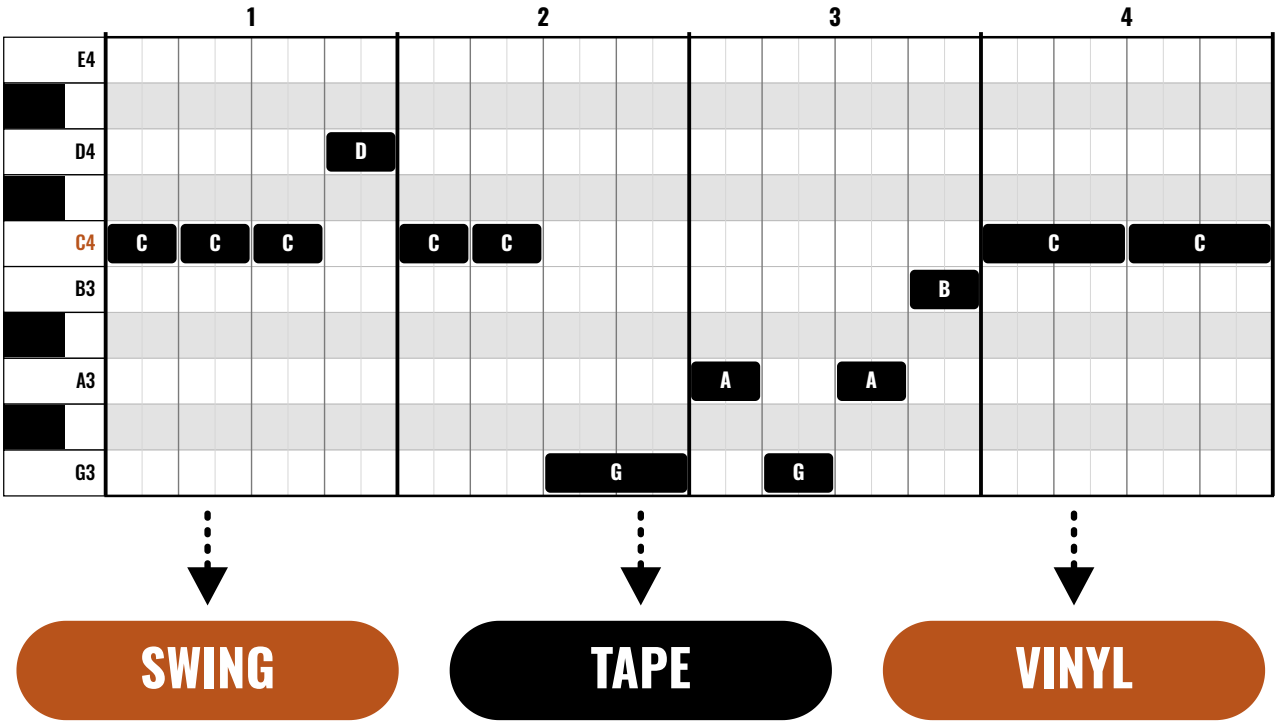
HUDSON BEAT



TURN A CAROL INTO LO-FI HIP-HOP

LO-FI HOLIDAY BEAT

Logic Pro · piano-roll charts · swing, tape, vinyl · intro slides

GRADES 9–12 · 4 CLASS PERIODS · 27 PAGES + 15-SLIDE DECK

LO-FI HOLIDAY BEAT

How to use this resource

START HERE

Teacher

WHAT THIS IS ·

A 4-class-period beat-making project for high school music technology. Each student chooses one of three public-domain carols, enters its melody and chords into Logic Pro from a piano-roll note chart, and turns it into a lo-fi hip-hop beat: a slow, swung groove with tape wobble, a Bitcrusher, a low-pass filter and vinyl noise. The beat is 60 to 90 seconds. Students plan it, build and mix it, present it and reflect on it. It is an original Hudson Beat project.

QUICK START ·

1. Read the pacing guide (page 5) and the teacher notes (pages 6–7).
2. Decide which carols to offer (page 9) and check the setup (page 4).
3. Print the student pages (pages 12–26), one set per student.
4. Open the intro deck and read the speaker notes for Class 1.

THE INTRO DECK ·

Every slide has a class: Class 1 (slides 1–6), Class 2 (slides 7–9), Class 3 (slides 10–11) and Class 4 (slides 12–15). Every slide has speaker notes.

Fonts: the deck uses Oswald and Source Sans 3 (free on Google Fonts, built into Google Slides). If PowerPoint or Keynote shows a different font, install the two fonts or present the PDF copy, which shows the intended look.

WHAT'S INSIDE ·

PART	PAGES	USE
Teacher pages	pages 2–11	How to use, overview, setup, pacing, teacher notes, Logic quick reference, tune sources, worked example (2 pages)
Student pages	pages 12–26	Three note charts, drum patterns, effect cards (2 pages), Beat plan, Arrangement plan, project sheet (3 pages), rubric, vocabulary, feedback planner, reflection page
Intro deck (separate file)	deck	15 slides, .pptx + PDF, speaker notes on every slide

NO AUDIO INCLUDED

No audio, recordings, lyrics, artists or albums are named or included. The three melodies are public domain and appear only as note charts. Students make every sound in their beats themselves, from the instruments, drum kits and plug-ins in Logic.

MAKE IT FIT YOUR CLASS ·

- **Easier or harder:** see the variations on page 6: one looped section on Good King Wenceslas, or a counter-melody.
- **Carol choice:** two winter or Christmas carols and a winter tune about an evergreen (page 9). Offer the ones that suit your school.
- **Feedback:** students write three questions and answer classmates' on a class board or on paper.
- **Grading:** the rubric, /25, the average of the student's and your score (page 23).

IN ONE SENTENCE

Students learn that a style is made of small choices: a slow tempo, swing, softened sounds and a little noise turn the same tune into lo-fi.

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LO-FI HOLIDAY BEAT

Project overview

OVERVIEW

Beat-making project

THE PROJECT IN ONE LINE

Lo-Fi Holiday Beat · Beat making · Logic Pro

Turn a carol everyone knows into a lo-fi hip-hop beat.

WHAT STUDENTS MAKE ·

One original beat, 60 to 90 seconds long, built on a public-domain carol. The melody is entered from a piano-roll chart and stays recognizable; the tempo is 70 to 90 BPM, the drums are swung, and the sounds are softened, wobbled and crushed with a layer of vinyl noise. It has at least five layers and an intro, beat sections and an outro, is mixed with volume, pan and EQ, and uses at least four of the five lo-fi moves. Students hand in a bounced .mp3 or .wav and the zipped .logicx project, plus their plans and a reflection, then play the beat and take questions.

WHAT THIS PROJECT REINFORCES ·

Teaches how a style is built from small choices. Students learn that a slow tempo, swing, softened and crushed sounds and a layer of noise, not only the notes, make a beat sound lo-fi, and they practice entering notes and chords in Logic Pro's Piano Roll.

STANDARDS ·

Supports the National Core Arts Standards for Music (Music Technology, high school): generating and developing ideas (MU:Cr1.1, MU:Cr2.1), presenting work (MU:Pr6.1) and responding to it (MU:Re7.2).

GOALS ·

- Enter a melody and chords into Logic Pro's Piano Roll from a note chart, with the right pitches and lengths
- Program a swung lo-fi drum pattern and a bass line at 70 to 90 BPM
- Use swing, tape, bitcrushing, a low-pass filter and vinyl noise on purpose, and explain what each one does
- Plan, build and mix a 60 to 90 second beat with an intro, beat sections, a break and an outro

AT A GLANCE ·

Grades	9–12, music technology or music production
Length	4 class periods: choose and enter, build the beat, make it lo-fi and arrange, present and reflect
Software	Logic Pro: Piano Roll, instruments, Step Sequencer or Drummer, Q-Swing, effects, bounce
You supply	Macs with Logic Pro, headphones, a place to hand in files and a class board or paper for feedback
Assessment	Rubric, 5 categories scored 1–5, /25; final grade = average of the student's and the teacher's score (page 23)

BEFORE THEY START

Students should know how to open a project, add a software instrument and use the mixer in Logic. Nobody needs to read music: the charts name every pitch and length.

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LO-FI HOLIDAY BEAT

Setup and what to watch

SETUP

Teacher

YOU NEED ·

- **A Mac with Logic Pro** for each student, and **headphones**. Speakers help for the final listen.
- **A place to hand in** the .mp3 or .wav, the zipped .logicx and the reflection.
- **The three charts**, printed (pages 12, 13 and 14), or shown on the projector.
- **The drum patterns and effect cards**, printed (pages 15–17).
- **Printed student pages**, one set each (pages 12–26).
- **A MIDI keyboard** is optional: every melody can be drawn with the mouse.

BEFORE CLASS 1 ·

- [ ] Decide which carols you will offer (page 9).
- [ ] Open Logic Pro on a student computer: can you make a track, open the Piano Roll and hear playback through headphones?
- [ ] Check that the plug-ins on page 16 and page 17 are installed (Bitcrusher, Channel EQ, Tape Delay) and that your Logic has a noise or crackle loop or a synthesizer with a noise oscillator.
- [ ] Read the worked example (pages 10–11).

WHAT TO WATCH ·

Students enter the notes but get a wrong one, the drums stay stiff, the melody is buried in noise, or the effects are on but nothing sounds lo-fi. Pairs well with a listening day: play the finished beats without saying the title and let the class name the carol. A tune that people name quickly is a beat that kept the melody.

IDEAS FOR THE ROOM

- Ask while students work: is it lazy yet? Can you hear the carol? Which lo-fi move is this?
- On Class 1, tap 70, 80 and 90 BPM on the desk so students feel how slow lo-fi is.
- A student who is stuck on drums can start from a Drummer track in a hip-hop style and swing it.

CHOOSING CAROLS ·

A RESPECTFUL CHOICE

Good King Wenceslas is a winter carol about snow and kindness, God Rest Ye Merry, Gentlemen is a Christmas carol, and O Tannenbaum is a winter tune about an evergreen tree. Offer the ones that fit your school and let students choose. If a student prefers another melody, a public-domain one you approve (first published in the United States in 1930 or earlier) works with the same plan, and the blank piano roll on page 18 is there for their own chart.

KEEP THE VOLUME MODERATE

Bass-heavy beats and loud peaks are tiring in headphones. Tell students to start low and check the final mix at a normal level.

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LO-FI HOLIDAY BEAT

Pacing guide · 4 class periods

PACING

Teacher

4 class periods. The “Plan” column is the day-by-day pacing; the other columns show what to put on the screen and in students’ hands.

CLASS	PLAN	TEACHER MOVES	DECK	STUDENT PAGES
CLASS 1	Choose a carol, fill in the Beat plan, and enter the melody and chords.	Slides 1–6: the hook, what the project is, the four classes, the five moves and how to read a piano roll. Hand out the project sheet, the three charts and the Beat plan. Check that each student has a carol and a tempo before they leave.	Slides 1–6	p. 12, p. 13, p. 14, p. 18, p. 20, p. 21, p. 22, p. 24
CLASS 2	Build the beat: drums, swing and a bass line.	Slides 7–9: building the beat, straight against swung, and three lo-fi drum patterns. Demonstrate the Step Sequencer and Q-Swing on the projector, then work time. Walk the room with one question: is it lazy yet?	Slides 7–9	p. 15
CLASS 3	Make it lo-fi, plan the form, and build the arrangement and mix.	Slides 10–11: the worked example on paper and the four sound moves. Hand out the Arrangement plan. Then work time; ask each student to play their first 20 seconds and say which carol they hear.	Slides 10–11	p. 16, p. 17, p. 19
CLASS 4	Bounce, hand in, present and reflect.	Slides 12–15: what to turn in, the rubric, presentation day and the close. Students bounce, name their files and write their three feedback questions, then present. Reflection (page 26) follows the presentations.	Slides 12–15	p. 23, p. 25, p. 26

THE ONE THING TO DO ON CLASS 1

Make sure every student leaves with a carol, a tempo from 70 to 90 BPM and four chords written on the Beat plan. If the tempo is only a guess, tap it: tap a steady beat for ten seconds and count the taps.

IF YOU ONLY HAVE THREE CLASSES

Merge Classes 2 and 3: build the beat and add the lo-fi effects on Class 2, then arrange and mix on Class 3 and use Class 4 for bouncing, presentations and reflection.

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WHAT THIS PROJECT REINFORCES ·

Teaches how a style is built from small choices. Students learn that a slow tempo, swing, softened and crushed sounds and a layer of noise, not only the notes, make a beat sound lo-fi, and they practice entering notes and chords in Logic Pro's Piano Roll.

COMMON STUDENT STRUGGLES ·

- The notes go in but the tune sounds wrong: check the grid setting and the note lengths against the chart, bar by bar (Logic quick reference, page 8).
- The drums sound stiff, not lazy: check that swing is on and that the melody sits a little behind the beat (page 16).
- The beat is muddy or buried in noise: turn the effects and the vinyl noise down until the carol is clear (page 17).
- Effects are on but nothing sounds lo-fi: compare with the cards on pages 16–17: a low-pass filter, a little wobble, grit on one sound and quiet noise under everything.

SUGGESTED VARIATIONS ·

Easier: Use Good King Wenceslas, loop one four-bar section for the whole beat, and start from pattern A on the drum patterns page or a Drummer track in a hip-hop style.

More Advanced: Write a second four-bar section with a different chord order, add a counter-melody or a chopped copy of the melody in Quick Sampler, and automate the filter across the whole piece. O Tannenbaum is the hardest chart: it is in 3/4, so it needs the waltz pattern.

WHERE THIS KIT HELPS WITH EACH STRUGGLE

- Wrong notes: the chart pages (pages 12, 13 and 14) list every note by bar, and the Logic quick reference (page 8) shows the grid.
- Stiff drums: the swing card (page 16) and slide 8 show what swing does.
- A muddy or noisy mix: the effect cards (page 17), the mix checks on the Arrangement plan (page 19) and slide 11.
- Effects with no lo-fi feel: the effect cards (pages 16–17) and the worked example (pages 10–11).

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SUGGESTED RESOURCES ·

RESOURCE	HOW TO USE IT
The three note charts (pages 12, 13 and 14)	Hand out one chart each, or let students choose. Pitch names are printed, so nobody needs to read staff notation.
Lo-fi drum patterns (page 15)	Students copy a pattern into the Step Sequencer or use a Drummer track. Pattern D is the 3/4 waltz for O Tannenbaum.
Lo-fi effect cards (pages 16–17)	One card each for swing, tape, bitcrushing, vinyl noise and the low-pass filter, with the setting to try and what to listen for.
Worked example (pages 10–11)	Good King Wenceslas as an 84-second lo-fi beat, planned on paper. Use it as a model, not a template; student projects will differ.

HOW TO GRADE ·

- The rubric: five categories, each scored 1–5, /25 (page 23). Students score themselves first; the final grade is the average of their score and yours.
- Melody & Chords (rubric row 1): open the project and compare the Piano Roll with the chart, bar by bar. Check that the chords match the plan.
- Lo-Fi Beat and Lo-Fi Sound (rubric rows 2 and 3): check the tempo is 70 to 90 BPM, that swing is on, and that at least four of the five lo-fi moves can be heard and named.
- Arrangement and Mix (rubric row 4): compare the project with the Arrangement plan (page 19). Is there an intro, beat sections, a break and an outro, and can you hear the melody over the noise?
- Plan and Reflection (rubric row 5): the Beat plan (page 18), the questions on the feedback planner and the Reflection page (page 26) are the evidence.

REFLECTION ·

Every student reflects (page 26): a written or oral conversation about how you did on the project and what you would do differently next time. Students write it or talk it through with you; either way they answer the four prompts. Allow about 10 minutes after the presentations. Use it as evidence for the Plan & Reflection row of the rubric.

NOTES / OBSERVATIONS ·

Pairs well with a listening day: play the finished beats without saying the title and let the class name the carol. A tune that people name quickly is a beat that kept the melody.

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LO-FI HOLIDAY BEAT

Logic quick reference

SETUP

Teacher + student

Where to find the tools a Lo-Fi Holiday Beat is likely to use. Demonstrate what your class needs on Classes 1 and 2 and then let students work. Names are from current Logic Pro versions; labels move between versions, so check yours.

New project and tracks	File › New…, then add a Software Instrument track with Track › New Tracks…. Use one track for the melody and more for chords, bass and drums.
Tempo and time signature	Type the tempo in the control bar (70 to 90 BPM). O Tannenbaum is in 3/4: set the time signature before you enter notes (the control bar, or the Signature track in Global Tracks).
Make a region, open the Piano Roll	With the Pencil tool, click on the track where bar 1 starts to make an empty MIDI region, then double-click it to open the Piano Roll editor.
Draw notes and chords	Pick a grid value in the Piano Roll's Snap menu (for example, 1/16 note) so notes land on the chart's sixteenth-note grid, click with the Pencil to add a note and drag its right edge to change its length. Draw a chord as notes stacked in one column. A MIDI keyboard is optional.
Fix wrong notes	Select a note and drag it up or down a row, or press delete. To play a melody an octave lower, select the region and change Transpose in the Region inspector by 12 steps.
Swing	Select the MIDI region. In the Region inspector, set Quantize to a 1/16-note value, then raise Q-Swing so the off-beat notes land late. The slider is Logic's own scale, so compare with the diagram on page 16 by ear.
Drums	Make a Drummer track and pick a hip-hop style, or make a Drum Machine Designer track and open the Step Sequencer: click the 16 steps for each drum, as on the patterns on page 15.
Bass, keys and sounds	Browse the Library (press Y) for instruments, or open Alchemy, Retro Synth or Vintage Electric Piano. A Session Player (Bass Player, Keyboard Player) can follow a Chord Track in Global Tracks.
Tape and bitcrush	Add plug-ins in a channel strip's Audio FX slots. For wobble try Scanner Vibrato or a light Chorus; for echo try Tape Delay with low feedback; for warmth try ChromaGlow if your version has it, or a little Overdrive. For grit use Bitcrusher.
Low-pass filter	In Channel EQ, switch on the High Cut filter (a low-pass filter) and slide it down until the highs soften. Draw its frequency with automation to open and close it.
Vinyl noise	Search the Loop Browser (press O) for a noise or crackle loop, or make quiet noise from a synthesizer's noise oscillator (Alchemy has one) and filter it with Channel EQ. Keep it very low in the mix.
Automation, mixing and bounce	Press A to show automation, pick a parameter such as a filter cutoff or volume, and draw points. Press X for the Mixer. Then File › Bounce › Project or Section (Command-B), MP3 or WAV. In Finder, Control-click the .logicx and choose Compress.

VERSION NOTE

Menu names are from current Logic Pro versions; older versions may differ slightly. Logic Pro is a trademark of Apple Inc., named only to describe what the project needs. This resource is not affiliated with Apple.

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LO-FI HOLIDAY BEAT

Tune sources and the public-domain rule

TEACHER

All three melodies are in the public domain, and each is printed here as a note chart only: no lyrics, no recordings and no published edition. The charts are written in C major or A minor (all white keys) so the pitches are easy to read, and checked against the notes in code.

TUNE	SOURCE · YEAR	IN THIS KIT	GOOD FOR
Good King Wenceslas the first phrase	melody "Tempus adest floridum" 1582	C major, 4/4 4 bars, 13 notes	The easiest chart: all quarter notes and two halves, one long note at the end of each line. A good first choice.
God Rest Ye Merry, Gentlemen the first line	English traditional melody early 1800s	A minor, 4/4 4 bars, 14 notes	A natural for lo-fi: a minor key, steady quarter notes and a long note in bar 4. Sounds moody with seventh chords.
O Tannenbaum (O Christmas Tree) the first line	traditional German tune 1820	C major, 3/4 4 bars, 15 notes	The one to pick for a waltz feel: it is in 3/4, so it takes the waltz drum pattern (pattern D). A challenge, and a winter tune with no holiday in its title.

WHERE THE TUNES COME FROM ·

- **Good King Wenceslas:** A 13th-century spring carol, first published in the Finnish collection Piae Cantiones in 1582. The English words (1853) are not used.
- **God Rest Ye Merry, Gentlemen:** An English traditional melody in print in the early 1800s (an 1815 piano rondo, a parody in the 1820s). The words are not used.
- **O Tannenbaum (O Christmas Tree):** A traditional German tune, printed in August Zarnack's school songbook in 1820. The words (1824) are not used.

THE PUBLIC-DOMAIN RULE ·

In 2026 a melody is in the public domain in the United States when it was first published in 1930 or earlier (1931 from 1 January 2027). If a student suggests another carol or winter song, check its first publication date before you approve it, and never use a recording, a film or pop version, or its lyrics unless they are public domain too.

PICKING A TUNE FOR A STUDENT ·

- **First time in the Piano Roll:** Good King Wenceslas. Mostly quarter notes, and a loop that repeats cleanly.
- **Wants a moody sound:** God Rest Ye Merry, Gentlemen. A minor key, so the seventh chords sound dark and warm.
- **Wants a challenge, or a waltz:** O Tannenbaum, in 3/4 with the waltz pattern.

READING THE CHARTS

Pitch names run down the left; middle C is C4. Bold lines are bars, grey lines are beats. A note's length is the name beside it: 16th, eighth, dotted eighth, quarter, dotted quarter, half or dotted half. Chords are one idea that fits, not the only one.

WHY A FOUR-BAR LOOP

Lo-fi beats are built on a short loop that repeats. Each chart ends so that it leads back into bar 1; where a tune starts with a lead-in note, that note sits at the end of bar 4.

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LO-FI HOLIDAY BEAT

Worked example · an invented beat

EXAMPLE

Teacher

An **invented plan on paper**: Good King Wenceslas, the four-bar loop, as a lo-fi beat at 80 BPM. No audio is included. Student projects will differ. Page 1 of 2: the plan and the lo-fi moves.

1 · A WEAK PLAN

I will make Good King Wenceslas into a chill lo-fi beat.

It names no tempo, no moves and nothing that would sound lo-fi.

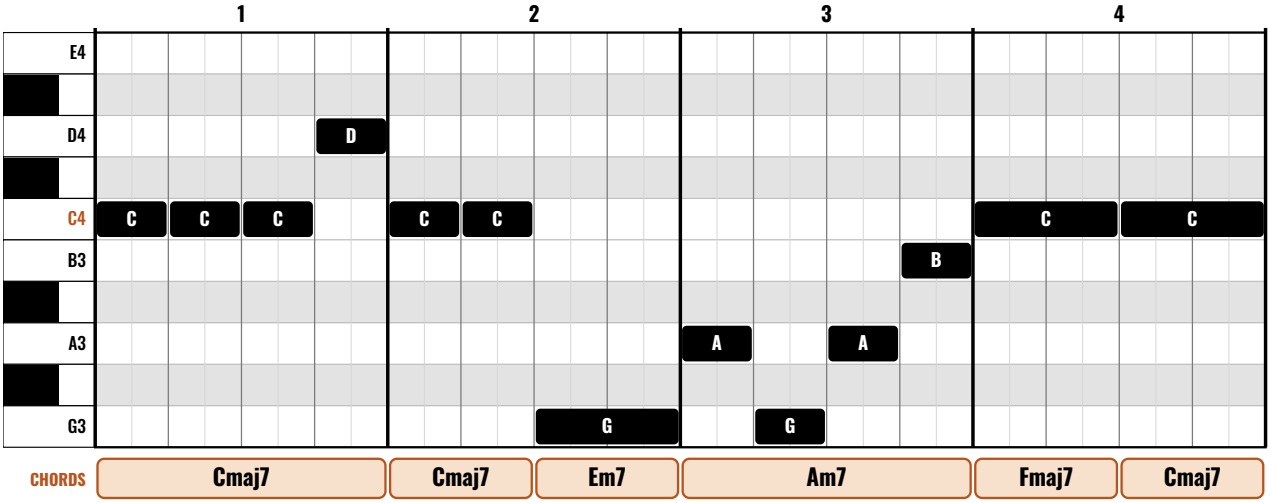
2 · A CLEAR PLAN

My carol is Good King Wenceslas, the four-bar melody from the chart in C major. My beat is lo-fi hip-hop at 80 BPM. The loop is 4 bars with the chords Cmaj7 | Cmaj7 Em7 | Am7 | Fmaj7 Cmaj7. I use all five lo-fi moves: swing on the hi-hats, tape wobble and echo on the keys, a Bitcrusher on the melody, a low-pass filter on the keys in the intro and the break, and vinyl noise under everything. The beat is 28 bars, 84 seconds: intro, beat, break, beat with bells, outro.

3 · FIVE LO-FI MOVES, AND WHAT STAYS ·

Swing	the off-beat hi-hat 16th notes land late (Quantize 1/16 and Q-Swing, page 16)
Tape	a little Scanner Vibrato for wobble and a Tape Delay with low feedback on the keys
Bitcrush	a Bitcrusher on the melody, with Mix about half
Low-pass	Channel EQ High Cut closes on the keys in the intro and the break
Vinyl noise	a quiet noise track under the whole beat
Stays	Every pitch and every length of the melody chart: the carol keeps its tune, so people still know it.

4 · THE LOOP ON THE PIANO ROLL, WITH ITS CHORDS (BARS 1–4) ·



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LO-FI HOLIDAY BEAT

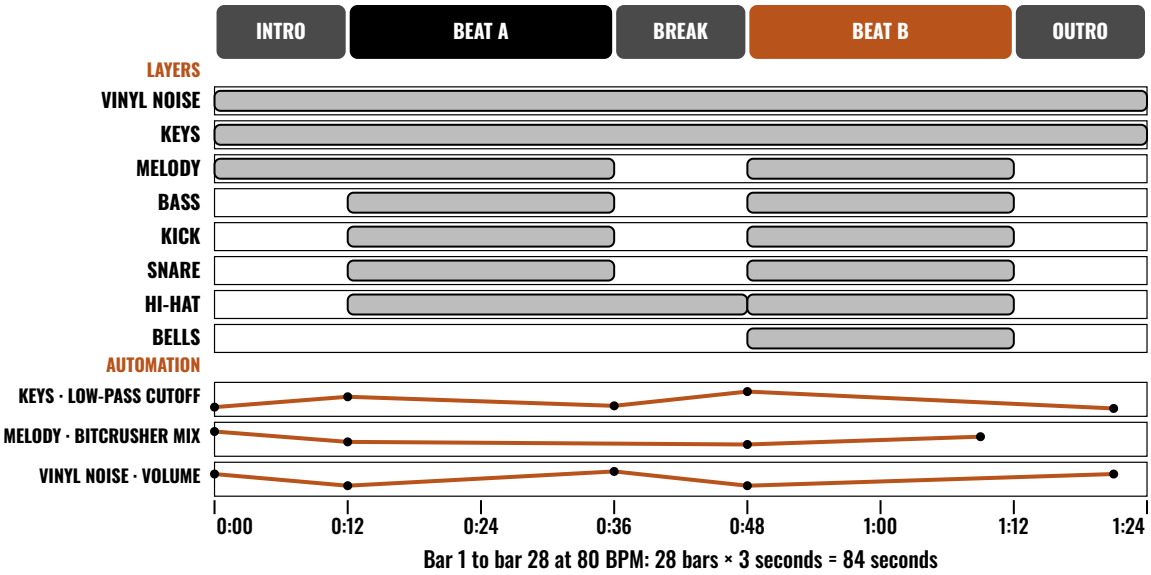
EXAMPLE

Worked example · the map

Teacher

The same beat, page 2 of 2: the map of the whole piece, then the numbers.

5 · THE MAP: 1:24, FIVE SECTIONS, EIGHT LAYERS, THREE LANES ·



Bar 1 to bar 28 at 80 BPM: 28 bars × 3 seconds = 84 seconds

6 · SECTIONS

Intro	bars 1–4	0:00–0:12
Beat A	bars 5–12	0:12–0:36
Break	bars 13–16	0:36–0:48
Beat B	bars 17–24	0:48–1:12
Outro	bars 25–28	1:12–1:24

7 · LAYERS (IN TO OUT)

Vinyl noise	a very quiet noise track under the whole beat, a little louder in the break
Keys	an electric piano playing the chords, with a low-pass filter closed in the intro and the break
Melody	the carol on a soft pluck sound with a Bitcrusher; it rests in the break
Bass	a soft sub bass: long notes on the chord roots (C2, E2, A1, F2)
Kick	pattern A: kick on beat 1 and just after beat 3
Snare	snare on beats 2 and 4
Hi-hat	swung eighth-note hats; they carry on through the break
Bells	the melody an octave higher on a bell sound, only in Beat B

8 · THE CHORDS OF THE 4-BAR LOOP, WITH THE BASS ROOT (THE MELODY'S FIRST AND LONGEST NOTES ARE CHORD TONES)

Bar 1	**Bar 2**	**Bar 3**	**Bar 4**
Cmaj7	Cmaj7 · Em7	Am7	Fmaj7 · Cmaj7
bass C2	bass C2 · E2	bass A1	bass F2 · C2

9 · CHECKING THE NUMBERS

One bar of 4/4 at 80 BPM lasts 240 ÷ 80 = 3 seconds, so 28 bars = 84 seconds, inside 60 to 90. The sections are 4 + 8 + 4 + 8 + 4 bars = 28. Beat B has the most layers (8). **Three feedback questions (invented):** 1. Which carol did you hear, and how fast did you know? 2. Where did the beat drop out, and what stayed? 3. Which lo-fi move did you notice first?

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LO-FI HOLIDAY BEAT

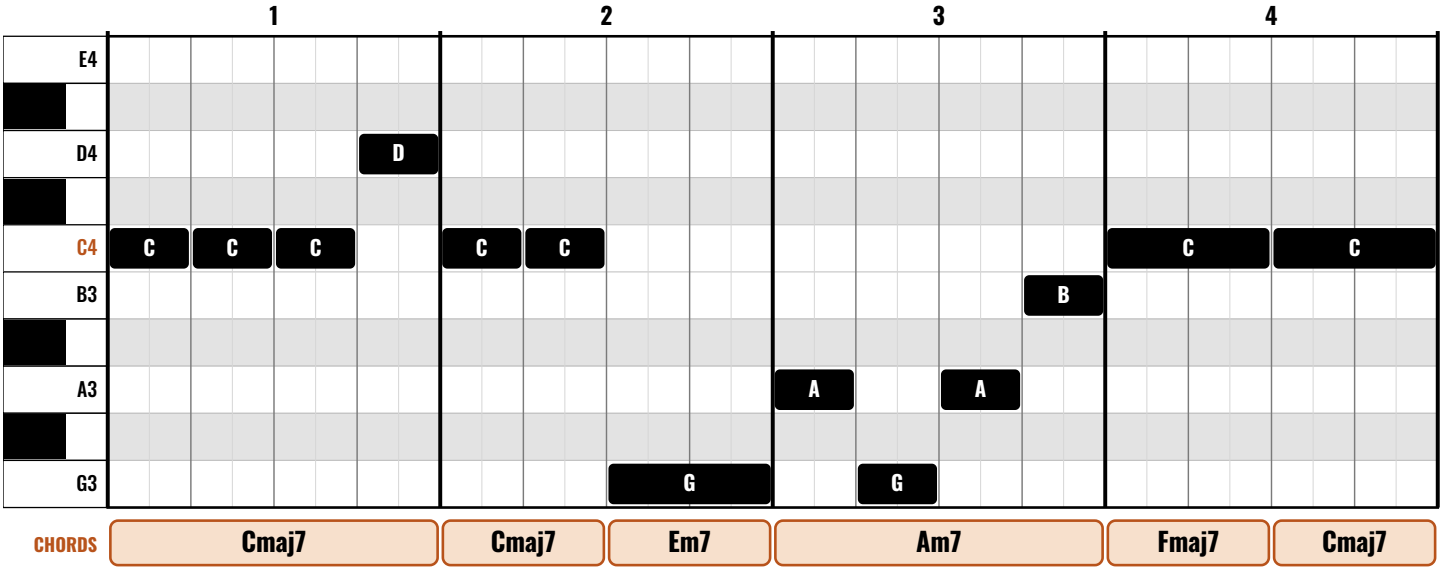
STUDENT

Note chart · Good King Wenceslas

Class 1

Name _____ Date _____

Good King Wenceslas, the first phrase · melody “Tempus adest floridum”, 1582 · C major, 4/4 · 4 bars · 13 notes

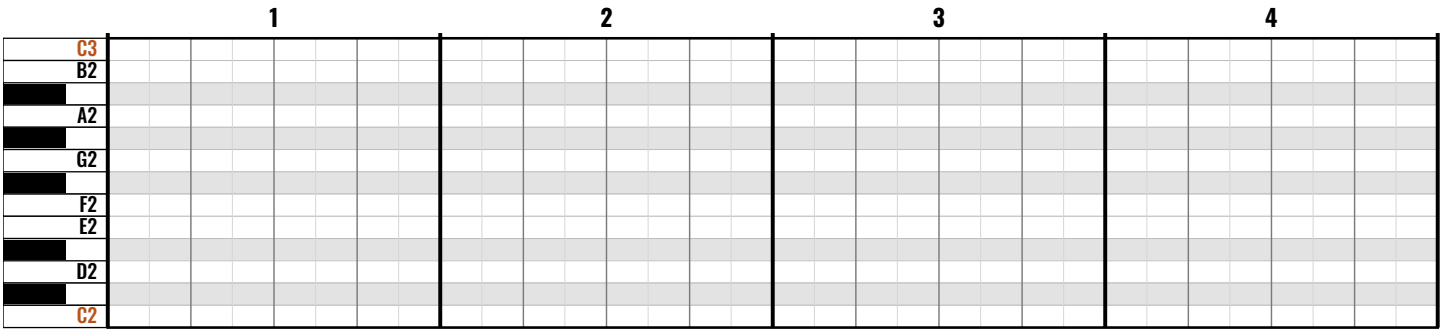


BAR	NOTES (PITCH, LENGTH)	CHORDS
1	C4 quarter · C4 quarter · C4 quarter · D4 quarter	Cmaj7
2	C4 quarter · C4 quarter · G3 half	Cmaj7 · Em7
3	A3 quarter · G3 quarter · A3 quarter · B3 quarter	Am7
4	C4 half · C4 half	Fmaj7 · Cmaj7

ABOUT THIS CHART

A 13th-century spring carol, first published in the Finnish collection Piae Cantiones in 1582. The English words (1853) are not used. Bar 1 is C C C D; bar 4 is two long Cs, so the loop breathes before it starts again. Loop all four bars. Chords shown are one idea that fits; try others by ear. Drums: pattern A, B or C (page 15).

YOUR TURN: WRITE A BASS LINE ON THE CHORD ROOTS (PENCIL), ONE LONG NOTE PER CHORD



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LO-FI HOLIDAY BEAT

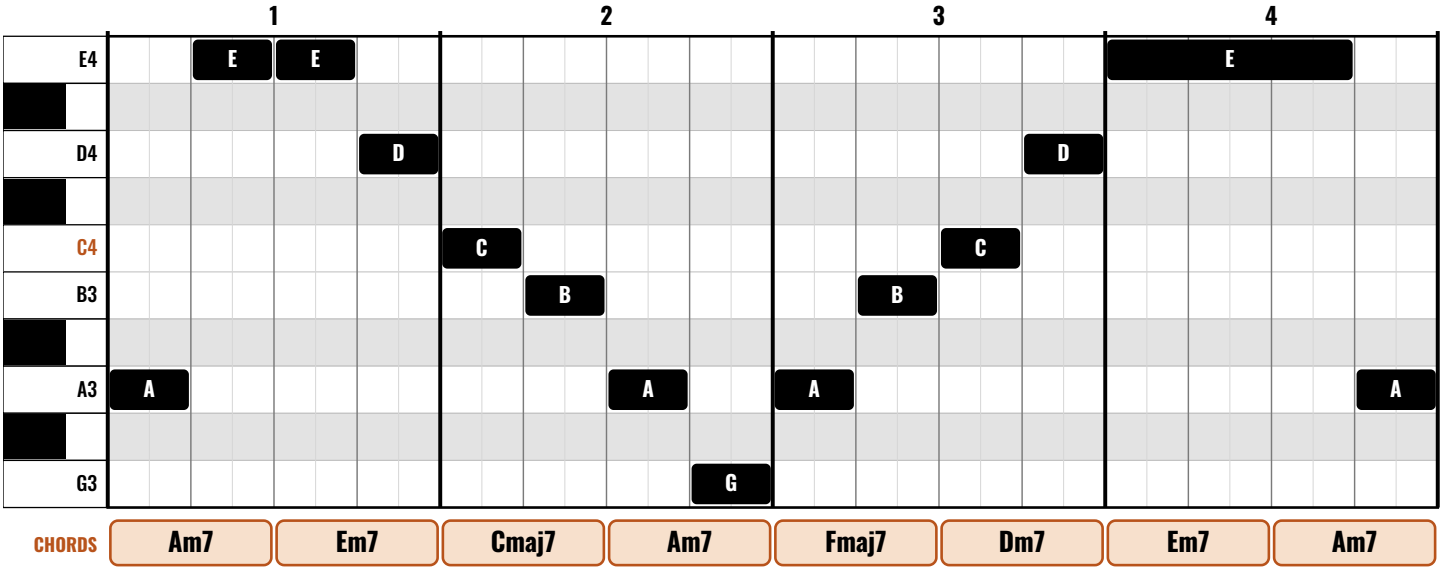
Note chart · God Rest Ye Merry, Gentlemen

STUDENT

Class 1

Name _____ Date _____

God Rest Ye Merry, Gentlemen, the first line · English traditional melody, early 1800s · A minor, 4/4 · 4 bars · 14 notes

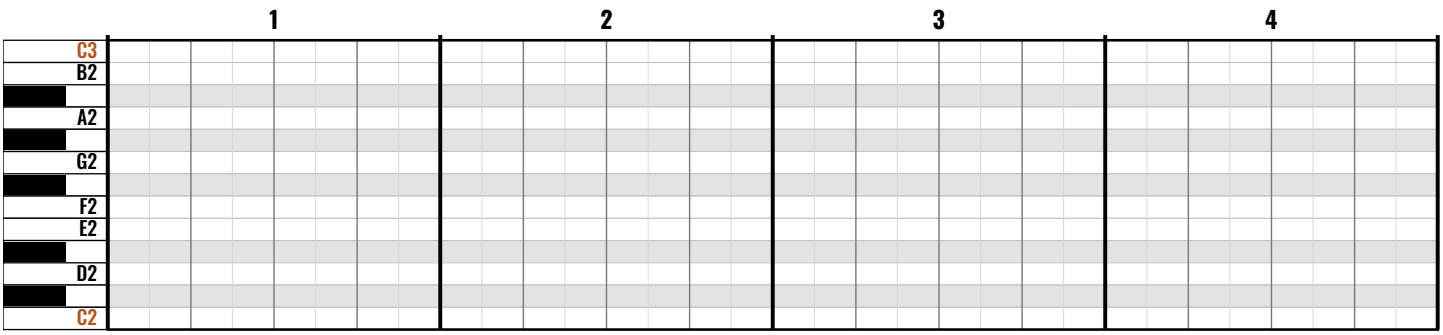


BAR	NOTES (PITCH, LENGTH)	CHORDS
1	A3 quarter · E4 quarter · E4 quarter · D4 quarter	Am7 · Em7
2	C4 quarter · B3 quarter · A3 quarter · G3 quarter	Cmaj7 · Am7
3	A3 quarter · B3 quarter · C4 quarter · D4 quarter	Fmaj7 · Dm7
4	E4 dotted half · A3 quarter	Em7 · Am7

ABOUT THIS CHART

An English traditional melody in print in the early 1800s (an 1815 piano rondo, a parody in the 1820s). The words are not used. The last quarter of bar 4 (A) is the lead-in to bar 1, so the loop turns around smoothly. Leave it in. Chords shown are one idea that fits; try others by ear. Drums: pattern A, B or C (page 15).

YOUR TURN: WRITE A BASS LINE ON THE CHORD ROOTS (PENCIL), ONE LONG NOTE PER CHORD



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LO-FI HOLIDAY BEAT

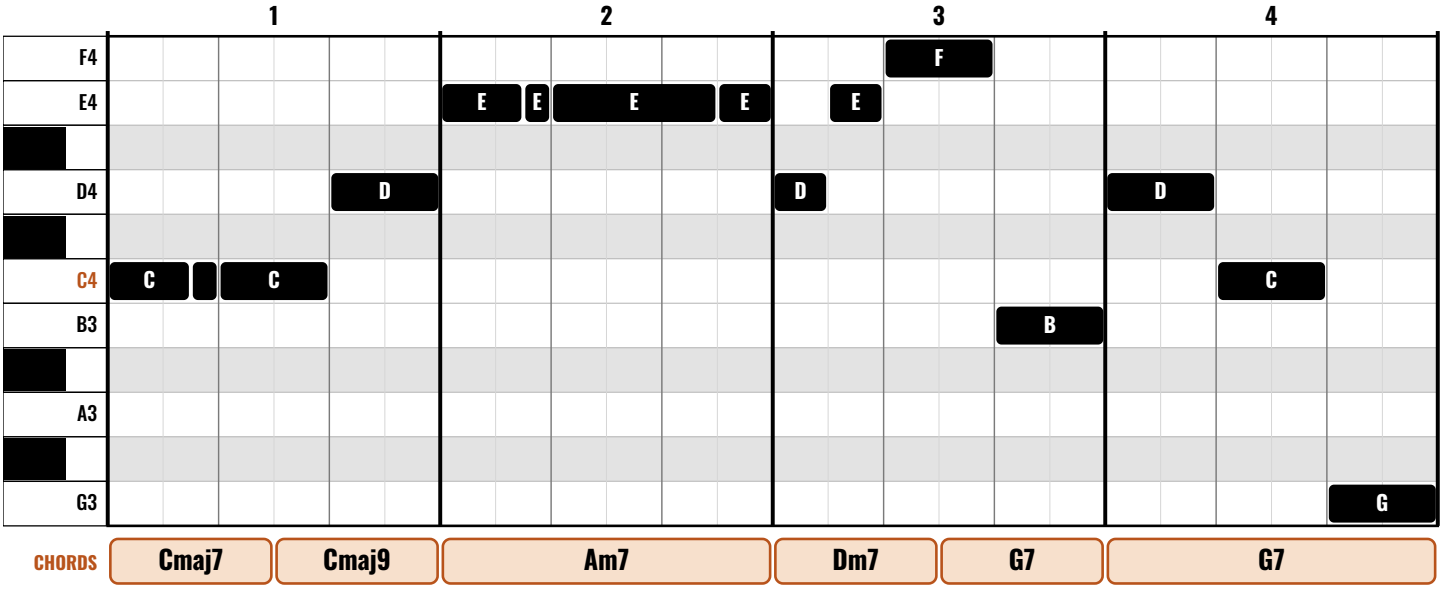
STUDENT

Note chart · O Tannenbaum (O Christmas Tree)

Class 1

Name _____ Date _____

O Tannenbaum (O Christmas Tree), the first line · traditional German tune, 1820 · C major, 3/4 · 4 bars · 15 notes

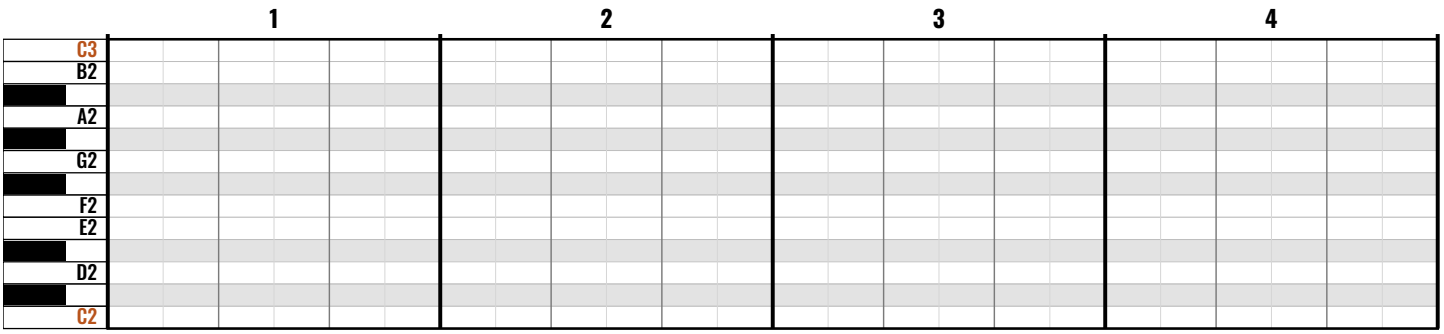


BAR	NOTES (PITCH, LENGTH)	CHORDS
1	C4 dotted eighth · C4 16th · C4 quarter · D4 quarter	Cmaj7 · Cmaj9
2	E4 dotted eighth · E4 16th · E4 dotted quarter · E4 eighth	Am7
3	D4 eighth · E4 eighth · F4 quarter · B3 quarter	Dm7 · G7
4	D4 quarter · C4 quarter · G3 quarter	G7

ABOUT THIS CHART

A traditional German tune, printed in August Zarnack's school songbook in 1820. The words (1824) are not used. Set the time signature to 3/4 first. Each bar is three beats: count 1-2-3. The last quarter of bar 4 (G) leads back into bar 1. Chords shown are one idea that fits; try others by ear. Drums: pattern D, the waltz (page 15).

YOUR TURN: WRITE A BASS LINE ON THE CHORD ROOTS (PENCIL), ONE LONG NOTE PER CHORD



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LO-FI HOLIDAY BEAT

Lo-fi drum patterns

STUDENT

Class 2

Lo-fi drums are simple and soft. Each card gives a one-bar pattern (16 steps, 4 to a beat; shaded steps are the beats; the waltz has 12), a bass idea and sounds to try. Pick one for your beat at 70 to 90 BPM, copy it into the Step Sequencer, or use a Drummer track, then swing it (page 16). Patterns A to C are for the 4/4 carols; pattern D is for O Tannenbaum.

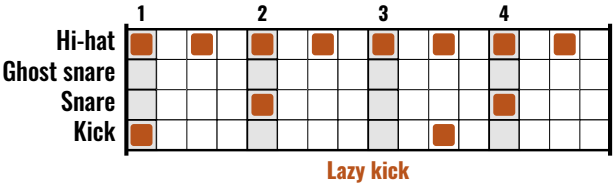
PATTERN A · LAZY KICK · RELAXED, WITH SPACE

Drums. Kick on beat 1 and an eighth note after beat 3; snare on beats 2 and 4; hats on every eighth note.

Bass. Soft sub: one long note on each chord root.

Sounds. A dusty kit with a soft kick, a short snare and quiet, swung hats.

Try. Turn the hats down so they are felt, not heard.



Lazy kick

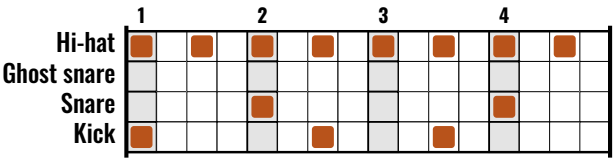
PATTERN B · BOOM-BAP · NODDING, A LITTLE BUSIER

Drums. Kick on beat 1, the “and” of 2 and the “and” of 3; snare on beats 2 and 4; hats on every eighth note.

Bass. Follow the kick: short notes on the root, a little behind the beat.

Sounds. A punchy but soft kick, a snare with a short tail, a touch of room reverb.

Try. Nudge the snare a hair late. Lo-fi drummers rarely sit dead on the grid.



Boom-bap

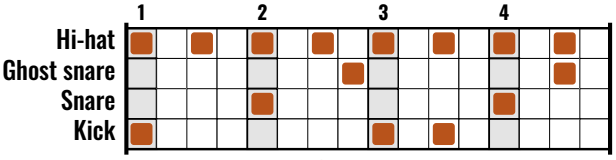
PATTERN C · GHOST NOTES · ROLLING, WITH LITTLE TAPS

Drums. Kick on beats 1 and 3 and an eighth note after beat 3; snare on 2 and 4; very quiet ghost snare hits on the last sixteenth of beat 2 and the “and” of beat 4; hats on every eighth.

Bass. Long roots, with a short walk-up note before each chord change.

Sounds. The same kit as A, with the ghost snare played at very low velocity.

Try. In the Piano Roll or Step Sequencer, set the ghost notes’ velocity low, about a quarter of full.



Ghost notes

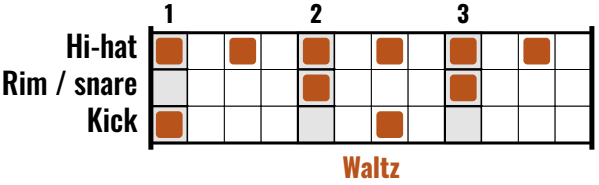
PATTERN D · WALTZ (3/4) · SLOW, ROCKING, FOR O TANNENBAUM

Drums. Kick on beat 1 and the “and” of 2; rim or snare on beats 2 and 3; hats on every eighth note. One bar is 12 steps.

Bass. One long root per bar.

Sounds. Soft kick, a rim click, quiet hats with swing.

Try. Count 1-2-3 out loud while you click the steps. Keep the swing light.



Waltz

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LO-FI HOLIDAY BEAT

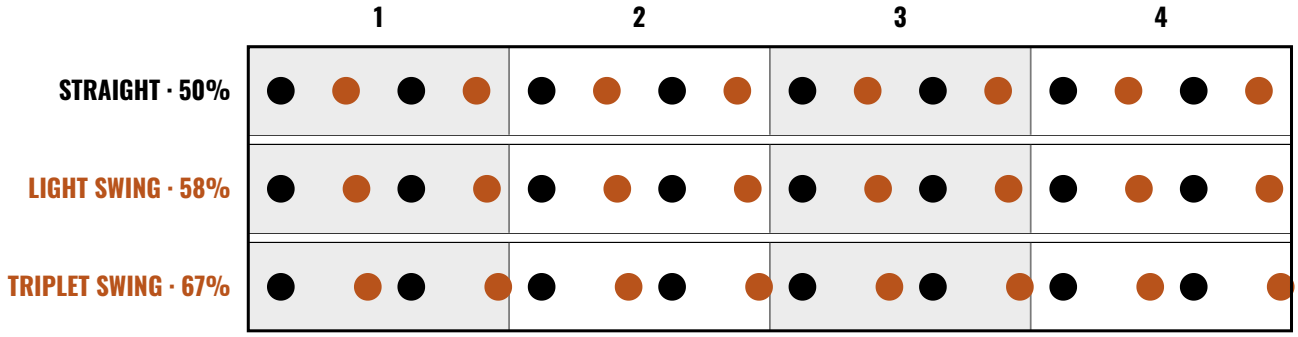
Lo-fi effect cards · 1 of 2

STUDENT

Class 3

Five small moves make a beat sound lo-fi: swing, tape, a Bitcrusher, a low-pass filter and vinyl noise. Use at least four. This page has swing and tape; page 17 has the other three. Every move should be small enough that the carol is still easy to hear.

SWING · MAKE THE BEAT BOUNCE



At 80 BPM a 16th note lasts 187.5 ms: light swing is 30 ms late, triplet swing 62.5 ms late

What it does. Moves the off-beat 16th notes a little later, so a steady pattern rolls and bounces instead of marching. More swing, more bounce.

In Logic. Select the MIDI region. In the Region inspector, set Quantize to a 1/16-note value, then raise Q-Swing. Try the hats first.

Try. Light swing on the hats, a little less on the kick and snare. Q-Swing is Logic's own scale, so use the diagram as a guide and decide by ear.

Listen for. The pattern leans back instead of marching. If it sounds loose or sloppy, you have too much.

TAPE · WOBBLE, WARMTH AND ECHO

What it does. Old tape drifts a little in pitch, softens the highs and adds warm distortion and echo. A touch of all of that makes a digital sound feel played from a tape.

In Logic. Put Scanner Vibrato or a light Chorus on the keys for a slow, small wobble. Add Tape Delay with low feedback for a soft echo. For warmth use ChromaGlow if your version has it; if not, turn Overdrive down to a tiny amount.

Try. Keep the wobble depth low: the pitch should drift, not warble. Set the delay to an eighth note and the mix low.

Listen for. A slightly seasick warmth. If the chords sound out of tune, back the depth off.

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LO-FI HOLIDAY BEAT

Lo-fi effect cards · 2 of 2

STUDENT

Class 3

BITCRUSHER · GRIT

What it does. Throws away detail: it lowers the bit depth and the sample rate, so a sound gets gritty and fuzzy, like an old sampler.

In Logic. Insert Bitcrusher in an Audio FX slot. Lower Resolution and Downsampling, then pull Mix back so some of the clean sound stays.

Try. Use it on one sound, such as the melody or the drums. Around 12 bits is subtle; around 8 bits is crunchy. Start with Mix near half.

Listen for. Grit you can hear on the tail of a note or on the hats. If the melody turns to mush, raise the bits.

VINYL NOISE · THE BED UNDER EVERYTHING

What it does. A soft bed of hiss and crackle, as if the beat were playing from a record. It glues the sounds together and says “lo-fi” at once.

In Logic. Search the Loop Browser (press O) for a noise or crackle loop, or make quiet noise with a synthesizer’s noise oscillator and filter it with Channel EQ. Put it on its own track and loop it across the whole beat.

Try. Turn it up until it is loud, then bring it down until you only notice it when it stops. Raise it a little in the break.

Listen for. Warmth under everything, not a wall of noise. The carol must stay clear.

LOW-PASS FILTER · MUFFLED AND FAR AWAY

What it does. A low-pass filter lets the low sounds through and cuts the highs, so keys and drums sound muffled, like a radio in another room.

In Logic. In Channel EQ, switch on the High Cut filter and slide it down. Automate its frequency to close it in the intro and the break and open it when the drums come in.

Try. A cutoff of about 3 to 6 kHz muffles keys; higher is a lighter touch.

Listen for. The highs soften and the sound sits back. If the keys go dull and lifeless, open the filter a little.

STARTING SETTINGS, ALL FIVE MOVES

Swing	Quantize 1/16, then raise Q-Swing until the hats bounce
Tape	Wobble depth low; Tape Delay feedback low, mix low
Bitcrusher	About 8 to 12 bits, Mix near half, on one sound
Low-pass filter	High Cut about 3 to 6 kHz on the keys
Vinyl noise	Quiet enough that you notice it only when it stops

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LO-FI HOLIDAY BEAT

Beat plan

STUDENT

Class 1

Name _____ Date _____

Step 2: plan. Pick one carol, a tempo from 70 to 90 BPM and a drum pattern, then decide which lo-fi moves you will use. You need at least four of the five (requirement 4), and the carol must still be recognizable.

MY CHOICES

My carol: _____ **Key and time signature:** _____

Tempo (BPM): _____ **Drum pattern (A, B, C or D):** _____ **Why this tempo:** _____

THE CHORDS OF MY LOOP (FROM MY CHART, OR MY OWN)

Bar 1	Bar 2	Bar 3	Bar 4

MY LO-FI MOVES (CHECK AT LEAST FOUR)

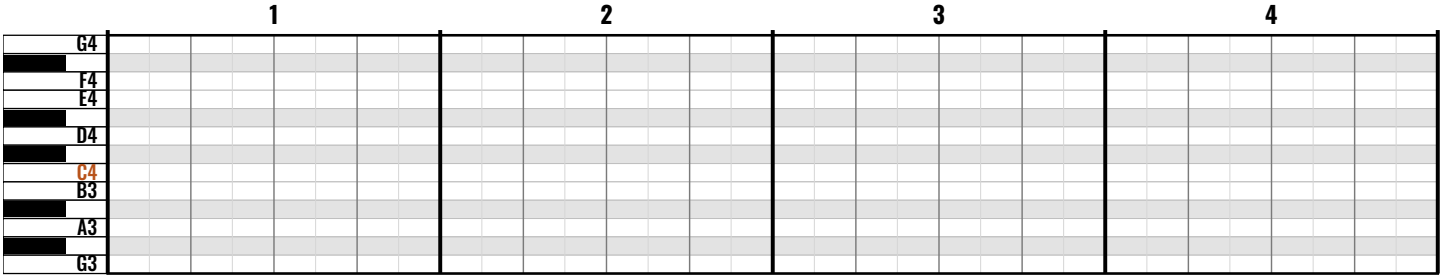
- ☐ **Swing** — where: _____
- ☐ **Tape wobble or echo** — where: _____
- ☐ **Bitcrusher** — where: _____
- ☐ **Low-pass filter** — where: _____
- ☐ **Vinyl noise** — where: _____
- ☐ **A soft bass on the chord roots** — where: _____
- ☐ **Ghost notes** — where: _____
- ☐ **Automation (filter or noise volume)** — where: _____

What stays the same, so people still know the tune: _____

MY LAYERS

LAYER	SOUND OR INSTRUMENT IN LOGIC	WHAT IT PLAYS

SKETCH: A COUNTER-MELODY OR A CHANGED RHYTHM FOR THE FIRST FOUR BARS (PENCIL, OPTIONAL)



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LO-FI HOLIDAY BEAT

Arrangement plan

STUDENT

Classes 3–4

Name _____ Date _____

Step 6: arrange. Give your beat a form: an intro, beat sections with a break, and an outro. Shade the bars where each layer plays, write the layer's name at the left, and mark where each section starts (every thick line is 4 bars). 60 to 90 seconds only.

Layer	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32

MY SECTIONS

Intro: _____ **bars** _____

Beat: _____ **bars** _____

Break: _____ **bars** _____

Outro: _____ **bars** _____

LENGTH HELPER (4/4)

BPM	ONE BAR	20 BARS	24 BARS	28 BARS
70	3.43 s	68.6 s	82.3 s	96.0 s
75	3.20 s	64.0 s	76.8 s	89.6 s
80	3.00 s	60.0 s	72.0 s	84.0 s
85	2.82 s	56.5 s	67.8 s	79.1 s
90	2.67 s	53.3 s	64.0 s	74.7 s

Aim for 60 to 90 seconds. Seconds = bars × 240 ÷ BPM. In 3/4, one bar is 180 ÷ BPM seconds.

My tempo: _____ **Bars:** _____ **Length: bars × 240 ÷ tempo =** _____ **seconds**

CHECKS BEFORE I BOUNCE

- ☐ I can name the carol in the first ten seconds, and the melody is louder than the drums and the noise.
- ☐ Every pitch and length matches the chart, or a change I planned.
- ☐ I can hear every layer, and none is buried or too loud.
- ☐ At least four lo-fi moves are on, each audible, and EQ is in use.
- ☐ The beat is between 60 and 90 seconds.

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Name _____ Date _____

LO-FI HOLIDAY BEAT

Music Technology | *Beat-Making Project* | 4 class periods

OVERVIEW ·

Turn a carol into a lo-fi hip-hop beat. Choose one of three public-domain melodies, enter its four-bar melody and chords into Logic Pro from a piano-roll note chart, build a slow, swung drum pattern, and make everything sound warm and a little old with tape wobble, a Bitcrusher, a low-pass filter and vinyl noise. Your finished beat is 60 to 90 seconds long at 70 to 90 BPM, and people should still be able to name the carol.

GOALS ·

- Enter a melody and chords into Logic Pro's Piano Roll from a note chart, with the right pitches and lengths
- Program a swung lo-fi drum pattern and a bass line at 70 to 90 BPM
- Use swing, tape, bitcrushing, a low-pass filter and vinyl noise on purpose, and explain what each one does
- Plan, build and mix a 60 to 90 second beat with an intro, beat sections, a break and an outro

KEY VOCABULARY ·

Lo-fi	Short for low fidelity: music made to sound a little rough, warm and old on purpose, with noise, wobble and muffled highs
Swing	A timing feel in which the off-beat notes land a little late, so a steady pattern bounces instead of marching
Bitcrusher	An effect that lowers a sound's bit depth and sample rate, adding a gritty, crunchy, digital roughness
Public domain	Music whose copyright has ended, so anyone may use the melody; the three carols in this project are

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LO-FI HOLIDAY BEAT

STUDENT

Project sheet · requirements and steps

Name _____ Date _____

REQUIREMENTS ·

- ☐ Choose one of the three carols (or a melody your teacher approves) and enter its four-bar melody into a Software Instrument track in Logic Pro; every pitch and length matches the chart
- ☐ Set a tempo from 70 to 90 BPM, build one of the lo-fi drum patterns (page 15) and swing it
- ☐ Play the chords of the loop under the melody, and add a bass line on the chord roots
- ☐ Use at least four of the five lo-fi moves: swing, tape, bitcrush, a low-pass filter and vinyl noise (pages 16–17)
- ☐ Build a 60 to 90 second arrangement with at least five layers, an intro, beat sections and an outro
- ☐ Reflect on how the beat went, in writing or out loud (page 26)

STEPS / PROCESS ·

1. **Choose** — Pick one carol and read its chords (carol charts: pages 12, 13 and 14)
2. **Plan** — Fill in your Beat plan: tempo, drum pattern, chords, and which lo-fi moves you will use (page 18)
3. **Enter the melody and chords** — Draw the chart into the Piano Roll and listen for wrong notes (Logic quick reference: page 8)
4. **Build the beat** — Program a drum pattern (page 15), swing it, and add a bass line on the chord roots
5. **Make it lo-fi** — Add tape wobble and echo, a Bitcrusher, a low-pass filter and vinyl noise (pages 16–17)
6. **Arrange and mix** — Plan the form on the Arrangement plan (page 19), build 60 to 90 seconds with an intro, beat sections and an outro, balance with volume, pan and EQ, and bounce with the file name format
7. **Present & get feedback** — Write three questions you want classmates to answer about your beat. Play it, say which lo-fi moves you used, and answer at least 2 of each classmate's questions. Then reflect (page 26).

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LO-FI HOLIDAY BEAT

STUDENT

Project sheet · tips and checklist

Name _____ Date _____

TIPS FOR SUCCESS ·

- Lo-fi is about restraint. Keep every effect small enough that the carol is still easy to hear.
- Play the melody softly and a little behind the beat. Lo-fi feels relaxed, never rushed.
- Add one effect at a time and listen before you add the next. If you cannot hear what an effect does, turn it up and then back down.
- Save a new version of your project before each big experiment, so you can always go back.

SUBMISSION CHECKLIST ·

- ☐ Bounced .mp3 or .wav handed in where your teacher says
- ☐ Zipped .logicx project file handed in
- ☐ Files named correctly: “Your Name – Lo-Fi Holiday Beat – Carol”
- ☐ Beat plan, Arrangement plan and Reflection page handed in
- ☐ Rubric self-score done (page 23)
- ☐ I can name the lo-fi moves I used and what each one does

NOTES ·

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LO-FI HOLIDAY BEAT

Rubric

RUBRIC

/ 25

Name _____ Date _____

GRADING RUBRIC /25

Lo-Fi Holiday Beat | Original Project

Each category is scored 1–5. Score yourself first; your final grade is the average of your self-score and your teacher's score.

Category	1	2	3	4	5
MELODY & CHORDS	The melody is missing or mostly wrong	Several wrong pitches or lengths; chords missing or clashing	Mostly right, with a few slips; chords fit most of the loop	Every pitch and length matches the chart; the chords fit	Exact, with chords and a bass line that support the melody; the carol is unmistakable
LO-FI BEAT	No clear drum pattern or tempo	A drum pattern, but stiff, or outside 70 to 90 BPM	A steady drum pattern in range, with some swing	A swung pattern with kick, snare and hats that fit the tempo	A relaxed, bouncing groove: swing, drums and bass lock together
LO-FI SOUND	No lo-fi effects	One or two effects, hard to hear or overdone	Four lo-fi moves used, each audible	Four or five moves, balanced, and the student can say what each does	Every effect is used on purpose, with automation, and the carol is still clear
ARRANGEMENT & MIX	No clear form; layers cover each other	A loop with little change; muddy mix	Intro, beat sections and outro; mostly balanced	Clear form with a break; layers enter and leave; clean mix	A form that builds and resolves; every layer audible, melody on top, noise quiet
PLAN & REFLECTION	Plans, presentation or reflection missing	Thin plan; little to say about choices	Plan and reflection complete, with basic detail	Clear plan; thoughtful reflection and good answers to questions	Insightful reflection and a polished, confident presentation

Student Score: _____ / 25	Teacher Score: _____ / 25
Final Grade = Average of Student + Teacher = _____ / 25	

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LO-FI HOLIDAY BEAT

Key vocabulary

STUDENT

Vocabulary

Name _____ Date _____

The four words from your project sheet, with definitions. Rewrite each in your own words, then show where it happens in your own project.

LO-FI — Short for low fidelity: music made to sound a little rough, warm and old on purpose, with noise, wobble and muffled highs

In my own words: _____

Where it shows up in my beat: _____

SWING — A timing feel in which the off-beat notes land a little late, so a steady pattern bounces instead of marching

In my own words: _____

Where it shows up in my beat: _____

BITCRUSHER — An effect that lowers a sound's bit depth and sample rate, adding a gritty, crunchy, digital roughness

In my own words: _____

Where it shows up in my beat: _____

PUBLIC DOMAIN — Music whose copyright has ended, so anyone may use the melody; the three carols in this project are

In my own words: _____

Where it shows up in my beat: _____

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LO-FI HOLIDAY BEAT

Present and get feedback

STUDENT

Classes 3–4

Name _____ Date _____

FROM YOUR PROJECT SHEET, STEP 7

Present & get feedback — Write three questions you want classmates to answer about your beat. Play it, say which lo-fi moves you used, and answer at least 2 of each classmate's questions. Then reflect (page 26).

MY THREE FEEDBACK QUESTIONS ·

Ask about things you really want to know, not yes-or-no questions.

Question 1: _____

Question 2: _____

Question 3: _____

CLASSMATES' QUESTIONS I ANSWERED (AT LEAST 2) ·

CLASSMATE	THEIR QUESTION	MY ANSWER

BEFORE I PRESENT ·

- ☐ Files named "Your Name – Lo-Fi Holiday Beat – Carol"; the .mp3 or .wav and zipped .logicx handed in
- ☐ Reflection page done (page 26)
- ☐ Three questions for classmates written
- ☐ Rubric self-score done (page 23)

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LO-FI HOLIDAY BEAT

Reflection · how did it go?

STUDENT

After presenting

Name _____ Date _____

WHAT A REFLECTION IS

A reflection is a written or oral conversation about how you did on the project and what you would do differently next time. Answer all four prompts in your own words. You choose how.

☐ **Written.** I will write my answers on this page.

☐ **Oral.** I will talk my answers through with my teacher. My teacher fills in the box at the bottom.

1 **What went well?** Name something in your project you are proud of, and say why it worked.

2 **What did you struggle with?** Say what was hard and what you tried to fix it.

3 **What would you do differently next time?** Be specific: a step, a choice, a sound or a tool.

4 **Look at the rubric (page 23).** Which category are you proudest of, and which would you work on first next time?

ORAL REFLECTION · TEACHER

Date: _____ **Prompts covered (1-4):** _____

Key points I heard: _____

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CREDITS

- Project sheet, rubric, teacher notes and intro deck: original work by Brian Lotze, Hudson Beat.
- Piano-roll note charts of three public-domain melodies (Good King Wenceslas, 1582; God Rest Ye Merry, Gentlemen, a traditional English melody; O Tannenbaum, 1820), the worked example, the drum patterns, the effect cards, the hero, card, swing and arrangement drawings and all layouts: original work, Hudson Beat. No lyrics, recordings or published editions are included.
- No audio, video, album art, screenshots, presets, templates, links or chart data are included. No student work is included.
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Created with the help of digital and AI tools. Notation and answer keys are verified before release.